

#UNIT-3

Quick Work

Page No.:

Date:

M T W T F S S

#Automated Data Mining

- 1) Refers to the process of using algorithms & computational methods to automatically extract knowledge, patterns & insights from large datasets without significant manual intervention
- 2) Approach leverages ML, AI & data mining techniques to streamline the analysis process, reduce human bias & accelerate decision-making

Basics of using a Modeler - SPSS Modeler

1) What is SPSS Modeler?

- Visual data mining & predictive analytics tool by IBM
- Allows non programmers to build models & analyze data using a node based drag & drop interface

- Applications:

- (i) Customer segmentation
- (ii) Fraud detection
- (iii) Risk scoring
- (iv) Forecasting trends
- (v) Predictive maintenance

2) Components -

< Already done in Unit-1 >

3) Reading data files -

- Common sources:

- (i) Excel files
- (ii) CSV files
- (iii) Databases (via ODBC)
- (iv) IBM Watson/cloud data
- (v) IBM SPSS statistics files (.sav)

#ADP-

Outputs-

- Preparation Nugget
- Report

Palette → Modeling → Auto Data Preparation

Typical operations

- | | |
|--------------------|-----------|
| 1) Imputation | 4) Bin |
| 2) Handle Outliers | 5) Filter |
| 3) Transform | |

- Steps:

- (i) Drag a source node to the canvas
- (ii) Double click the node & browse and select your dataset
- (iii) Use the type node (optional) to assign field roles

KDD process -

<Already done in Unit-1>

4) Data Exploration (EDA) -

- Critical to understand the characteristics of your dataset
- Useful nodes:
 - (i) Table
 - (ii) Distribution
 - (iii) Histogram
 - (iv) Statistics
 - (v) Matrix: Cross tabulations for pairs of field

5) Data Preparation -

- Key nodes:

- (i) Select
- (ii) Filter
- (iii) Type
- (iv) Replacing missing values
- (v) Derive
- (vi) Balance
- (vii) Reclassify

6) Building models -

- Modeling nodes:

- (i) Decision Tree
- (ii) Logic Regression
- (iii) Neural Net
- (iv) SVM
- (v) Random Forest
- (vi) Auto Classifier (automated model selection)

- Steps:

- (i) Connect the prepared data streams to the model node
- (ii) Configure the model
- (iii) Run the model to train it
- (iv) Evaluate using an analysis or evaluation node

7) Model evaluation -

- Nodes for evaluation:

- (i) Analysis
- (ii) Evaluation
- (iii) Histogram
- (iv) Table

8) Exporting or deploying models -

- Export model options:

- (i) Export model as PMML (Predictive Model Markup Language)
- (ii) Save model stream for reuse

- (iii) Export results to excel, csv, or databases

- Deployment options:

- (i) Schedule streams for automation
- (ii) Integrate with IBM Watson Studio or cloud services
- (iii) Use the export node for production ready outputs

Data Partitioning

Creates Partition Field

1 = Training

2 = Testing

3 = Validation (optional)

Data Partitioning -

- Involves dividing the validation & test set

* Place before any data transformation.
needed to prevent target leakage

Common patterns

(i) Standard (balanced)
70/30 (train/test)

(ii) 3 way split
60/20/20
(Train/Test/Valid)

(or)

* # Predictor Selection for

- Involves choosing which to include in the model based on their relevance

& impact on target variable

- Often include feature selection techniques such as

(i) Filter methods (eg. correlation analysis)

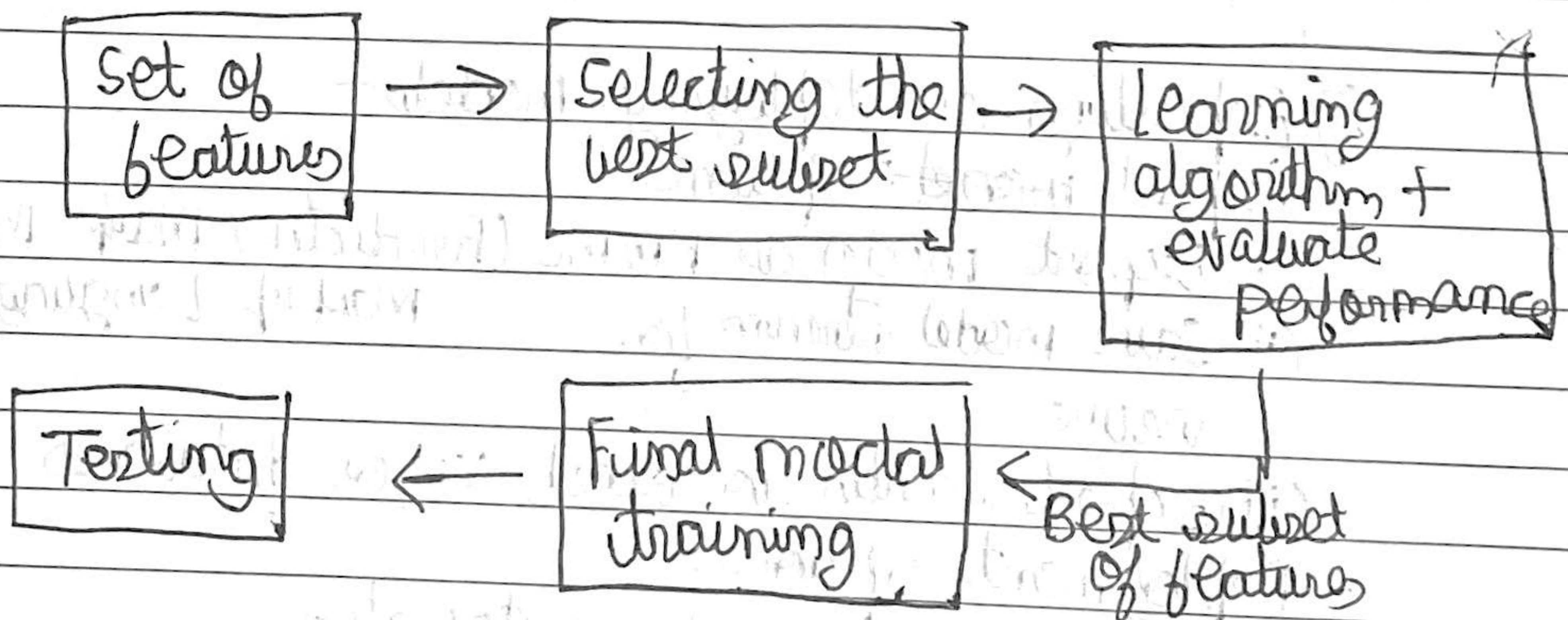
(ii) Wrapper methods (eg. recursive feature elimination)

(iii) Embedded methods (eg. regularization)

1) Filter methods -

- Process of selecting a subset of relevant features (variables, attributes) from a larger set of available set of features to build models

- Specifically focused on selecting features based on their statistical properties & scores



eg. Correlation analysis:

Measures the relationship between 2 or more variables

Range: -1 and $+1$

$+1$ = perfect positive

-1 = perfect negative

0 = no linear relationship

Purpose: Helps identify data patterns, predict relationships & support decision making



2) Wrapper methods:

- Feature selection techniques that evaluate subsets of features by training models repeatedly
- Features are selected based on their impact on model performance, not just statistical scores
- Unlike filter methods they test different combinations of features using actual model results
- Types: Can use forward (add features) or backward elimination (remove features)
- eg - Recursive Feature Elimination (RFE)
Iteratively removes the least important features to find the most significant ones

Process:

- Train a machine learning model
- Rank features by importance
- Remove the least imp features
- Retrain the model & repeat until the desired no. of features remains

3) Embedded methods -

- Combines the advantage of filter & wrapper methods by performing feature selection during model training
- Feature selection is built into the learning algorithm itself
- Benefit: Model learns both - the relationship between features & the target variable, & which features are most important
- eg- Lasso regression & decision tree

Regularization = used to prevent overfitting in ML models

Purposes = adds a penalty to large feature coefficients, reducing their magnitude toward zero

Effect = helps the model generalize better by controlling complexity

Goal = achieve a balance between underfitting (too simple) and overfitting (too complex)

Techniques of regularization -

(i) Ridge regression

(ii) Lasso regression

Ridge regression -

- Linear regression technique that adds a small bias to improve prediction stability
- Type: It uses L^2 regularization, adding a square of coefficients as a penalty term
- Purpose: reduces model complexity & prevents overfitting
- Advantages: Works well when the no. of features is greater than the no. of samples

Lasso Regression -

- Lasso regression (least absolute shrinkage & selection operations)

is a regularization technique used to reduce model complexity

- Type: It uses L1 regularization absolute value of coeff term

- Effect: Shrinks some coefficients

- Purpose: Helps in reducing the most ~~few~~ important

What automated node actually does?

- 1) Reads metadata
- 2) Build classification model
- 3) Score test/validation partition
- 4) Evaluation (Accuracy, Recall, Precision)
- 5) Rank them & return

Best model as model nugget
optional set up
evaluation report

Automated models for categorical

- These models predict outcomes that belong to discrete categories

- Eg: used in applications like customer churn, sentiment analysis & classification tasks

- Techniques: include decision trees, Logistic regression, Random Forests, Support Vector Machines, Neural Networks, LDA & RDA

- Goal: Automate classification by identifying patterns that separate distinct classes effectively

- Applications:

In healthcare, predicting patient outcomes (eg disease diagnoses) based on symptoms ~~involving~~ involves building models with categorical targets

Model evaluation -

- Measures how well a predictive model performs

- Metrics: Accuracy, Precision, Recall, F1 Score

- Purpose: To check the model's ability to generalize on new or unseen data

Source → Type → Partition → Automated (categorical) Modeling

Source → Type → Partition → Automated Modeling (continuous)

Quick Work

Page No.:

Date:

- ## # Automated models for continuous
- Automated models predict numerical values like house prices, stock prices, etc.
 - Use regression techniques:
 - (i) Linear Regression
 - (ii) Support Vector Regression
 - (iii) Ensemble Regression method
 - Purpose: Find relationships between continuous inputs & continuous outputs
 - eg - In energy demand forecasting, models predict electricity consumption using past usage data
- ## # What automated node actually does?
- 1) Reads metadata
 - 2) Builds regression models
 - 3) Scores test/validation partitions
 - 4) Evaluation (RMSE, MAE, R^2)
 - 5) Ranks them
 - 6) Outputs
 - Champion model nugget
 - Report comparing runners up

- ## # Deploying models -
- Involves integrating trained model into operational systems for real time or batch predictions
 - Models can be exported as APIs, scripts or executable files for use in production
 - Purpose: Make predictions available to end users on applications automatically
 - eg - A fraud detection model in banking uses APIs to score transactions and detect fraudulent activities in real time

Model Nugget -

Contains -

- (i) Trained model
- (ii) Remember which field it needs
- (iii) Prediction fields

* You don't retrain to deploy - you reuse the NUGGET

Deployment in SPSS -

New data → (same prep)

→ Model Nugget
→ Output / Export

Export / Save

- (i) File ⇒ Add Export node
- (ii) DB ⇒ Source → Prep
→ Model Nugget
→ Database Export

Essentials in the modeler -

- 1) Stream = a flowchart of nodes (data $\rightarrow$ prep $\rightarrow$ modeling
(output $\leftarrow$ evaluation) $\leftarrow$
- 2) Node = an operation (eg. read file, type, derive, evaluation)
- 3) Link = the connection that passes data/results between nodes

Important nodes -

- | | |
|---|---|
| 1) Source: <ul style="list-style-type: none"> (i) Van file (ii) Statistics file (iii) Excel (iv) Database | 2) Structuring & Filtering (Record Ops) <ul style="list-style-type: none"> (i) Select (ii) Sample (iii) Sort (iv) Aggregate |
|---|---|

- | | |
|--|--|
| 3) Field management (Field Ops): <ul style="list-style-type: none"> (i) Type (ii) Reclassify (iii) Replacing Missing values (iv) Partition (v) Set to Flag (vi) Derive | 4) Exploration & QA <ul style="list-style-type: none"> (i) Data audit (ii) Distribution/Graph (iii) Table |
|--|--|

5) Modeling & Scoring:

- (i) Classification: Logistic regression, SVM, Random Forest
 - (ii) Clustering: K-Means, Two Step
 - (iii) Association/Sequences: Apriori Sequence
 - (iv) Time series: Exponential smoothing, ARIMA
- * Each modeling node produces a model nugget to run

6) Evaluation & Output:

- (i) Evaluation: confusion matrix, lift, precision/recall
- (ii) Analysis: variable importance, gains, rule lists
- (iii) Output: table, export, database, excel

Super node

< Already done in unit-1 >

Model Evaluation:

1) Standard evaluation flow:

Source → Type → Partition → Prep (ADP/Derive)

→ [One or more Models] → Evaluation → Table / Export

2) Classification:

~~(i)~~ Confusion Matrix

- Precision

- Recall / Sensitivity

- Specificity

- FI

- Accuracy

UNIT-1

Evaluation

Node = Graphs / Output

~~(ii) ROC~~

3) Regression / Prediction:

Metrics:

- RMSE

- MAE

- R²

- MSE

Unit 2

of FML

Evaluation
NodeEvaluation
Node

4) Clustering / Segmentation:

K-Means

Two Step

DBSCAN

Unit 3

of FML

5) Association Rules:

Node

↳ Apriori

Unit 3

of FML

Support Lift Confidence

6) Time series forecasting:
Exponential smoothing / ARIMA

Methods:

└ MAE / RMSE

└ MASE

└ MAPE / SMAPE